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The effect of ambient temperature on cardiovascular mortality in 27 Brazilian cities

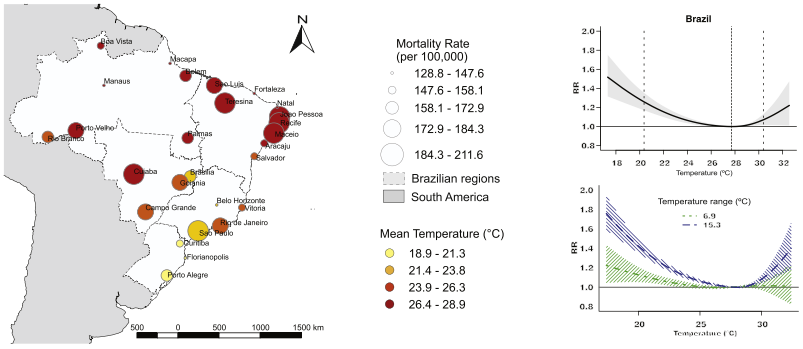
Ismael Henrique Silveira *, Beatriz Fátima Alves Oliveira, Taísa Rodrigues Cortes, Washington Leite Junger

Institute of Social Medicine, State University of Rio de Janeiro, Rua São Francisco Xavier, 524, Sala 7013-D, Maracanã, Rio de Janeiro Cep: 20550-013, RJ, Brazil

HIGHLIGHTS

- There is few evidence of the temperature effect on CVD mortality in Latin America.
- Temperature effects on CVD mortality were observed in most of the Brazilian regions.
- The association between temperature and daily CVD mortality in Brazil was U-shaped.
- The effects of temperature were stronger in places with a broad temperature range.

GRAPHICAL ABSTRACT



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ABSTRACT

Background: There is limited evidence on the relationship between temperature and cardiovascular mortality in middle and low-income countries, particularly in Latin America. In this study, we investigated the total effect of temperature on cardiovascular mortality in 27 Brazilian cities, and the effect modification by geographic, socioeconomic, demographic and infrastructure characteristics within cities.

Methods: In the city-specific analysis, we used time-series analyses to estimate the relationship between mean temperature and daily cardiovascular mortality using quasi-Poisson generalized linear models combined with distributed lag non-linear models. In the second stage, a meta-analysis was used to pool the effects of temperature on cardiovascular mortality for Brazil and its five regions (Central-West, North, Northeast, South, and Southeast). We used a meta-regression to examine the effect modification of city-specific geographic, socioeconomic, demographic and infrastructure-related variables.

Results: The risks associated with temperature varied across the locations. Higher cardiovascular mortality was associated with low and high temperatures in most of the cities, Brazil and the Central-West, North, South, and Southeast regions. The overall relative risk (RR) for Brazil was 1.26 (95% confidence interval [CI]: 1.17–1.35) for the 1st percentile of temperature and 1.07 (95% CI: 1.01–1.13) for the 99th percentile of temperature versus the 79th percentile (27.7 °C), where RR was lowest. The temperature range was the variable that best explained the variation in effects among the cities, with greater effects in locations having a broader range.

Conclusions: The results indicate effects of low and high temperatures on the risk of cardiovascular mortality in most of Brazil's capital cities, besides a pooled effect for Brazil and the Central-West, North, South, and Southeast

* Corresponding author.
E-mail address: ismaelhsilveira@gmail.com (I.H. Silveira).

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regions. These findings can help inform public policies addressing the health impact of temperature extremes, especially in the context of climate change.

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1. Introduction

Many epidemiological studies have identified an association between temperature and morbimortality due to CVD (Bai et al., 2016; Phung et al., 2016; J. Yang et al., 2015; Zhang et al., 2014; Zhou et al., 2017), and with all-cause mortality (Guo et al., 2014). The exposure-response relationship typically exhibits a J, U or V-shape and differs across study locations. The minimum mortality temperature was lower in colder places and higher in hotter locations (McMichael et al., 2008; Moghadamnia et al., 2017; Zhang et al., 2014).

The effects of ambient temperature on cardiovascular morbimortality can vary according to individuals characteristics such as age, socioeconomic status, sex (Bell et al., 2008; Ingole et al., 2017; Mari-Dell'Olmo et al., 2018; Son et al., 2016; J. Yang et al., 2015; Yu et al., 2010; Zhou et al., 2017), occupation (Ingole et al., 2017) and pre-existing cardiovascular diseases or other comorbidities (Lavigne et al., 2014). Evidence of effect heterogeneity was also observed according to geographic, socioeconomic, demographic, and infrastructure characteristics of the place of residence (Benmarhnia et al., 2015; Hajat and Kosatky, 2010; Moghadamnia et al., 2017) and between urban and rural areas (Urban et al., 2014; Wang et al., 2018).

Cardiovascular diseases (CVD) are the leading cause of death worldwide, accounting for approximately 32% of deaths in 2015. Over three-quarters of these deaths occurred in middle and low-income countries. Although the rates of mortality due to CVD have declined, particularly in high-income countries, the absolute number of deaths has increased in most nations (WHO, 2017). The increase has been attributed, besides an aging population, to difficulties in prevention and control of risk factors for CVD (Mendis, 2017; WHO, 2017). In addition to individual risk factors, previous studies have highlighted the role of environmental factors as determinants of CVD, such as ambient temperature (Bhatnagar, 2017; Diez Roux et al., 2016). However, the evidence of the effects of temperature on cardiovascular morbidity and mortality is predominantly for high-income countries with a temperate climate, whereas limited evidence is available for low-income countries with a tropical climate (Moghadamnia et al., 2017; Phung et al., 2016; Turner et al., 2012) such as Latin America and the Caribbean.

Brazil is the world's fifth largest country in terms of land area and sixth in population, and situated in a region of the globe subject to the greatest warming (IPCC, 2013). In 2015, around 354,000 deaths were recorded in Brazil, and CVD was the cause of 28% of this total (WHO, 2016). However, there is no previous multicity study in Brazil investigating the effect of temperature on cardiovascular mortality (Bell et al., 2008; Gouveia et al., 2003; Hajat et al., 2005; McMichael et al., 2008; Pinheiro et al., 2014; Son et al., 2016). Thus, the objective of this study was to investigate the total effect of ambient temperature on cardiovascular mortality in 27 Brazilian cities. We also explored the effect modification by geographic, socioeconomic, demographics and infrastructure characteristics within cities.

2. Materials and methods

2.1. Study design and data

We carried out a time-series study of 27 Brazilian cities, encompassing the capitals of each State and the Federal District. The study period spanned from 2000 to 2015, except for the cities of Curitiba, Florianópolis, Fortaleza, Porto Alegre and Teresina (2002–2015), and Rio de Janeiro (2003 to 2015).

Brazil can be divided into five main regions (Central-West, Northeast, North, South, and Southeast), 26 states and the Federal District (Fig. 1). The latitudes of the cities included in the study extend from 30.03 S in Porto Alegre to 2.82 N in Boa Vista. According to data from the 2010 census, São Paulo is the most populous city in the country with 11,253,503 inhabitants, while Palmas is the least populous with 228,332 (Table 1). The total population of the cities included in the study is 45,466,045, representing around 24% of the country's population (Atlas of Human Development in Brazil, n.d.).

We calculated the number of daily deaths due to CVD based on data from the Brazilian Mortality Information System (SIM) of the Information Technology Department of the Public Health Care System (DATASUS). We selected all deaths whose primary underlying cause was classified as in Chapter IX of the 10th Revision of the International Classification of Diseases (ICD-10).

The daily meteorological variables, mean temperature and mean relative humidity, were obtained from the automated meteorological stations of the Brazilian Institute of Meteorology (INMET), except for data of the cities of Campo Grande and Porto Velho. The data for these two cities were drawn from the airport meteorological data held by the Center for Weather Forecasting and Climate Studies (CPTEC) of the National Institute for Space Research (INPE).

The socioeconomic and demographic variables, including total population, percentage urban population, Gini index, percentage unemployment in the population aged 18 or more, Municipal Human Development Index (M-HDI), average income, and the percentage living in extreme poverty were obtained from the Atlas of Human Development in Brazil. The primary health care coverage and percentage of domiciles with trees were obtained from the DATASUS and the 2010 Demographic Census data, respectively.

2.2. City-specific analysis

On the time-series analysis, we used generalized linear models (GLM) with quasi-Poisson distribution combined with distributed lag non-linear models (DLNM). The DLNM, described by Armstrong (2006) and Gasparrini et al. (2010), and implemented in the *dlnm* library (Gasparrini, 2011), for the R program, allows the simultaneous representation of the non-linear dependence between exposure and response and lag effects. DLNM are implemented based on the definition of a cross-basis function, derived by combining two functions that describe the nature of the relationship throughout exposure and the lags. Thus, two basis functions are defined: one for the predictor and the other for lag, both of which will be combined.

Missing data were imputed using a method based on the EM algorithm implemented in the *mtsdi* library (Junger and Ponce de Leon, 2015) for the R program. The imputation used mean daily values for the meteorological variables temperature and relative humidity obtained from the ERA-Interim model of the European Centre for Medium-Range Weather Forecasts (ECMWF). Mean daily relative humidity was estimated based on the mean air temperature and mean temperature of the dew point using the *HeatStress* library for the R application (Casanueva, n.d.). The temporal pattern was modeled using generalized additive models (GAM), fitted using a natural cubic spline of the time variable with 4 degrees of freedom per year and an indicator variable for days of the week. The percentage of data imputed in each series of temperature and humidity data are shown in Table S1 of the Supplementary material.

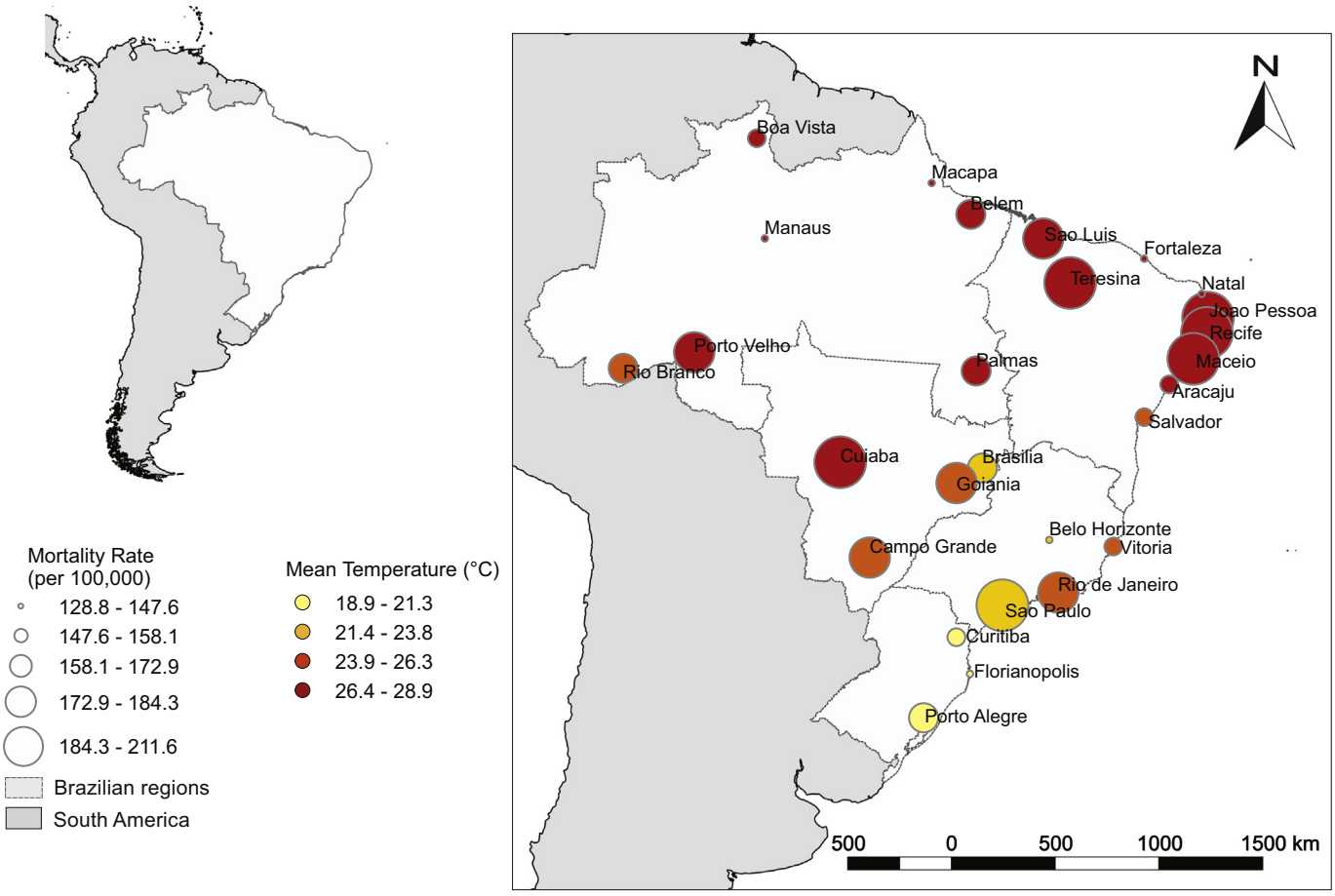


Fig. 1. Location of the 27 Brazilian capitals.

The variables were selected for the fitted models based upon a causal directed acyclic graph (DAG) (Pearl, 1995), depicted in Fig. S2 of the Supplementary material. Seasonality and long-term trends are adjusted in time-series studies because they influence patterns for both exposure and outcome (Bhaskaran et al., 2013). Day of the week was also included in the DAG as a common cause of both exposure and outcome for its potential influence on general aspects of life (Bhaskaran et al., 2013). Weather variables such as temperature and humidity have seasonal patterns, and cardiovascular mortality is higher in the winter (Fares, 2013). Air pollution can also be affected by weather conditions and follows seasonal patterns (Chen et al., 2015; Jhun et al., 2015). Concentrations of atmospheric pollutants, as well as their health effects, tend to be higher during colder and drier periods, although varying according to the pollutant. Humidity is related to temperature and thermal sensation, and may influence thermoregulation and consequently, the occurrence of cardiovascular outcomes (Davis et al., 2016; Liu et al., 2015). We used DAGitty (Textor et al., 2011) to select a minimal sufficient adjustment set of variables that would allow the identification of the total effect of temperature on CVD mortality: days of the week, seasonality and humidity. As depicted in the DAG in Fig. S2, air pollution was considered a mediator in the relationship between temperature and cardiovascular mortality as opposed to a confounding variable (Buckley et al., 2014), and was not fitted in the model. The equation below represents the fitted models for each location:

$$\log[E(Y_t)] = \alpha + \beta \mathrm{Temp}_{t,l} + \mathrm{ns}(\mathrm{RH}_t, 3) + \mathrm{ns}(\mathrm{time}, \mathrm{df} * \mathrm{years}) + \lambda \mathrm{DOW}_t$$

where Y_t is the daily death count due to CVD on days t of the series; α is the intercept; $Temp_{t,\ l}$ is the matrix of basis variables obtained by transformation of temperature during the lags; β is the vector of coefficients for $Temp_{t,\ l}$; $ns(RH_t, 3)$ is a natural cubic spline of relative humidity with three degrees of freedom; $ns(time, df * years)$ is the cubic spline of time, with number of degrees of freedom differing among cities; DOW_t is the variable indicating day of the week and λ is the vector of coefficients for DOW_t.

We modeled the non-linear lag effect of temperature using a cross-basis defined by a natural cubic spline for temperature, with three internal knots placed at the 10th, 75th and 90th percentiles, and cubic spline for the lags, with three equally spaced knots along its logarithmic scale and maximum lag of 21 days. The choice of spline to model the exposure-response relationship, as well as the allocation of the knots, was based on the minimization of Akaike's information criteria (AIC). We used maximum lag of 21 days to capture the effects of cold and heat, and also to adjust for possible harvesting effects (Bai et al., 2016; Chen et al., 2018; Gasparrini et al., 2015; J. Yang et al., 2015).

A natural cubic spline with 2 degrees of freedom was used to control for seasonality and long-term trends, except for the cities of São Paulo, Rio de Janeiro and Porto Alegre, for which 4 degrees of freedom were used. The number of degrees of freedom was defined based on examination of the periodogram and the autocorrelation plot, to prevent overfitting.

The effects of cold and heat extremes on cardiovascular mortality were calculated. These effects were defined by relative risks, cumulated during the 21-day lag period, at 1st and 99th percentiles of local mean temperature versus minimum mortality temperature (MMT), as adopted in other studies (Guo et al., 2014; Vicedo-Cabrera et al., 2016; Wang et al., 2018).

We performed several sensitivity analyses by varying the spline function, knots placement, maximum lag, number of knots for

Table 1
Population size, standardized mortality ratio and summary statistics for daily cardiovascular death, mean temperature, and relative humidity in 27 Brazilian cities.
SMR: Standardized mortality coefficient.

Region/cities	Population	SMR (/100,000 pop.)	Daily deaths			Temperature (°C)			Relative humidity (%)		
			Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
Central-West											
Brasília	2,570,160	161.9	7.6	0.0	18.0	22.6	15.0	30.6	61.9	15.0	97.0
Campo Grande	786,797	183.1	3.5	0.0	15.0	24.4	6.7	35.8	64.7	18.8	100.0
Cuiabá	551,098	198.8	2.0	0.0	9.0	27.6	11.9	36.0	70.6	27.0	96.3
Goiânia	1,302,001	178.2	4.9	0.0	15.0	26.0	16.2	34.9	59.0	19.7	94.3
North											
Belém	1,393,399	159.8	4.9	0.0	16.0	28.0	23.1	32.2	81.5	59.0	97.7
Boa Vista	284,313	152.7	0.6	0.0	5.0	28.7	23.3	34.5	73.0	30.0	99.5
Macapá	398,204	128.8	0.7	0.0	5.0	28.4	23.5	32.0	78.3	59.0	97.3
Manaus	1,802,014	146.2	3.8	0.0	12.0	28.2	21.5	34.5	80.6	45.3	99.3
Palmas	228,332	167.3	0.4	0.0	4.0	27.1	22.1	32.9	66.9	22.0	98.8
Porto Velho	428,527	173.6	1.1	0.0	6.0	26.6	15.0	35.5	82.5	25.0	100.0
Rio Branco	336,038	161.3	0.8	0.0	7.0	26.1	12.4	32.2	84.0	43.7	100.0
Northeast											
Aracaju	571,149	153.0	2.0	0.0	10.0	27.2	22.4	30.7	75.7	56.7	98.0
Fortaleza	2,452,185	146.3	8.1	0.0	23.0	27.9	23.8	31.1	74.9	53.0	96.3
João Pessoa	723,515	185.1	3.1	0.0	11.0	27.7	22.2	30.5	75.3	55.0	98.0
Maceió	932,748	211.6	4.1	0.0	14.0	26.5	21.7	31.0	78.4	50.5	130.7
Natal	803,739	145.7	3.1	0.0	14.0	27.3	22.7	30.3	79.8	64.0	98.0
Recife	1,537,704	184.6	8.0	0.0	22.0	27.0	22.0	30.5	75.7	48.0	98.0
Salvador	2,675,656	153.9	9.6	1.0	27.0	26.3	20.4	31.1	80.6	59.5	99.3
São Luís	1,014,837	182.0	3.3	0.0	11.0	28.0	23.6	31.8	80.0	55.3	98.0
Teresina	814,230	202.8	3.6	0.0	12.0	28.9	23.0	34.6	69.8	35.3	98.5
South											
Curitiba	1,751,907	157.0	7.4	0.0	22.0	18.9	3.8	29.0	78.5	34.7	99.0
Florianópolis	421,240	140.7	1.5	0.0	9.0	22.1	9.9	32.4	77.5	41.7	98.0
Porto Alegre	1,409,351	171.9	8.6	0.0	27.0	21.0	6.3	34.7	75.4	38.3	98.3
Southeast											
Belo Horizonte	2,375,151	130.7	9.6	0.0	27.0	23.1	14.8	31.5	61.2	22.7	97.3
Rio de Janeiro	6,320,446	181.6	39.9	17.0	89.0	26.0	15.5	35.0	71.1	39.0	93.0
São Paulo	11,253,503	184.9	59.0	29.0	106.0	21.4	7.9	32.1	71.2	24.3	99.0
Vitória	327,801	155.2	1.2	0.0	8.0	25.5	16.9	31.7	74.8	52.3	97.5

Note: Population from 2010 Census. Specific CVD mortality ratio, standardized for sex and age, is for 2009–2010 period, with 2010 population as base. Remaining information pertains to the study period of each city.

temperature and lag, number of degrees of freedom for fitting the temporal trend a model without the mean relative humidity variable.

2.3. Meta-analysis and meta-regression

The effects of temperature on cardiovascular mortality in each city were pooled to yield the mean effect on each of the five regions and Brazil. The meta-analysis models were used for multi-parameter nonlinear associations (Gasparrini et al., 2012; Gasparrini and Armstrong, 2013), implemented in the *mvmeta* library for the R program.

We performed a univariate meta-regression to examine the heterogeneity in the temperature-mortality relationship according to geographic, socioeconomic, demographic and infrastructure variables The following meta-variables were investigated: total population, percentage urban population, mean daily deaths due to CVD, Gini index, percentage unemployment in the population aged 18 or more, Municipal Human Development Index (M-HDI), average income, percentage living in extreme poverty, average minimum, mean and maximum daily temperatures, mean annual temperature range, latitude, primary health care coverage and percentage of domiciles with trees.

All analyses were carried out using the R program, version 3.4.2. A significance level of 5% was adopted for all statistical tests.

3. Results

Cardiovascular mortality rates, standardized by sex and age for the 2009–2011 period, ranged from 130.7 (per 100,000 population) to 211.6 (per 100,000 population). Averages of the daily mean temperatures were higher in the North (27.6 °C) and Northeast regions (27.4 °C). The Northeast region also had a lower mean temperature range (8.9 °C), while the North region had higher mean relative humidity (78.1%). The South region had the lowest mean temperature, high mean relative humidity (77.1%) and a broad temperature range (32.0 °C). The Central-West region was characterized by high daily mean temperature (25.2 °C), broad temperature range (21.9 °C) and low relative humidity (64.1%). The Southeast region had a moderate mean temperature (24 °C) and mean relative humidity (69.6%), and a broad temperature range (18.8 °C) (Table 1).

For most cities, we observed a clearly defined U-shaped exposure-response relationship and the minimum mortality temperature was generally higher in hotter places and lower in colder locations. Other capitals had poorly defined patterns. Belém, Macapá, Rio Branco and Teresina had a minimum mortality percentile close to 0, with an increased risk above this value. Aracaju, João Pessoa, Maceió, Natal, Recife, Salvador, São Luís, Florianópolis and Boa Vista had a minimum mortality percentile close to 1, and increasing risk below this value. Palmas had a W-shaped curve (Fig. S3 of the Supplementary material).

The pooled exposure-response curves for Brazil and the Central-West, South, and Southeast regions (Fig. 2) also exhibited a U-shape. The curve for the Northeast region showed a greater effect at very low temperatures and an apparently protective effect at high temperatures. In general, the cold effect was greater than the heat effect. On a national level, the effect of cold (1st percentile vs. MMP) increased the RR by 26% (95% CI: 17–35), while the effect of heat (99th percentile vs. MMP) increased the RR by 7% (95% CI: 1–13). The South region experienced the greatest cold effect, increasing the RR by 71% (95%CI: 1–13). The

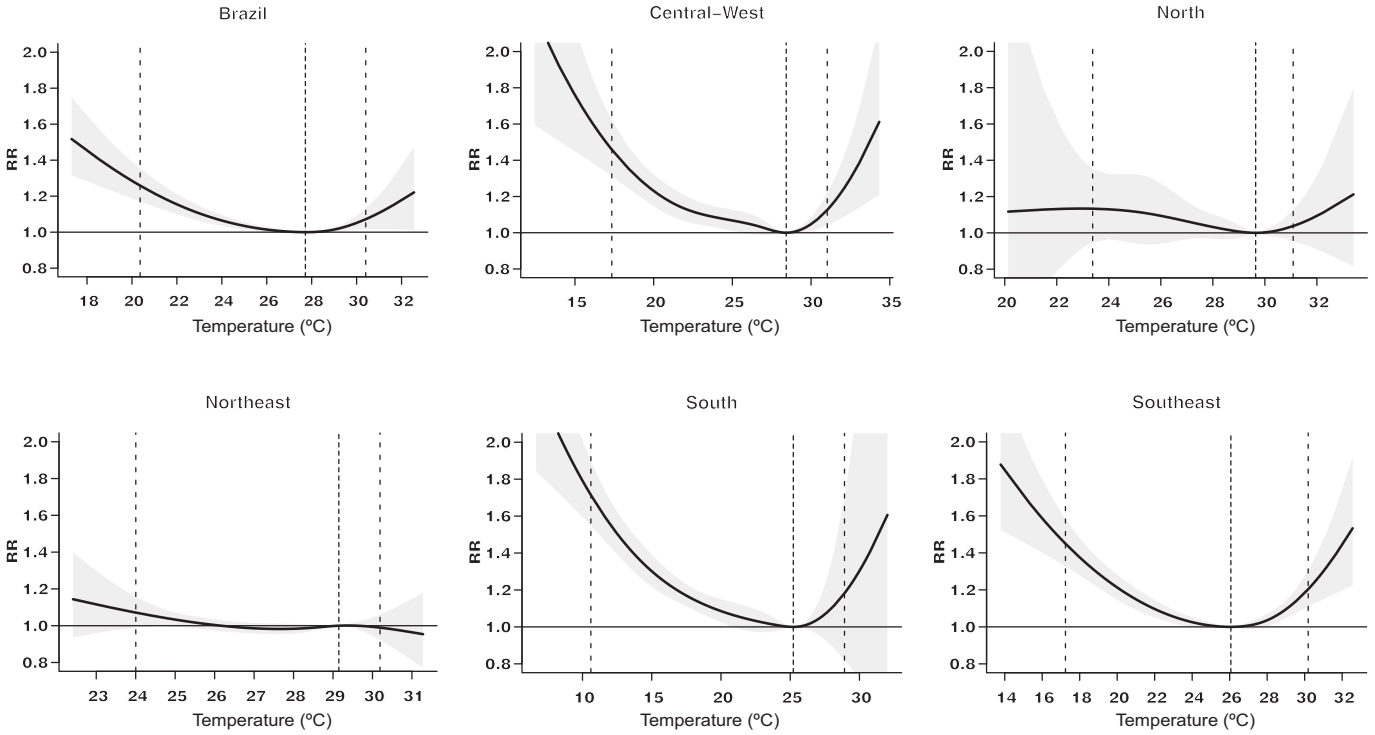


Fig. 2. Pooled overall cumulative exposure-response relationship between temperature and cardiovascular mortality over a 21-day lag period in Brazil and its five regions. The shaded grey areas represent the 95% confidence intervals, the dashed lines are the 1st and 99th percentiles of daily mean temperature and the dotted lines are the minimum-mortality temperatures.

Southeast region displayed the greatest heat effect, increasing RR by 20% (95%CI: 11–31). The minimum mortality temperature for Brazil was 27.7 °C (79th percentile), ranging from 25.2 °C (85th percentile) in the South region to 29.6 °C (90° percentile) in the North region (Table 2).

The pooled heat effect for Brazil was immediate, extending up to approximately the fifth day of lag, whereas the cold effect was highest at around the second day of lag, extending up to the third week. In Central-West and Northeast regions, the heat effect was not immediate, peaking at around the second day. The curve for the Northeast region was the most atypical and the effect for both cold and heat increased again after the tenth lag day. Although not significant, there is some evidence of harvesting effects due to heat over a longer lag period for the Central-West and Southeast regions and over a shorter lag period for the Northeast and South regions. Harvesting effects due to cold could have occurred at around the fifth day of lag in the Northeast region (Fig. 3).

Table 3 shows the I^2 statistic (measuring heterogeneity of the relationship between temperature and cardiovascular mortality in the cities analyzed), p-value of Cochran's Q-test for heterogeneity, p-value of Wald's test for the association with the meta-predictor variables and the AIC of the model.

Temperature range seems to explain a great part of heterogeneity between cities, with an I^2 equal 53.6% compared with the 76.3% of the model with no meta-variable. Other variables were also explanatory, such as minimum temperature, percentage living in extreme poverty, latitude, mean temperature, percentage unemployment, average income, human development index, maximum temperature, in this order. The meta-variables proved highly correlated with one another and the effects of temperature were more marked in places with high temperature range, lower mean and minimum and high maximum temperatures, lower latitudes and better socioeconomic conditions (Fig. 4). Descriptive statistics of the meta-variables investigated and Pearson's correlation coefficients are given in Tables S4 and S5 of the Supplementary material, respectively.

On the sensitivity analysis (Table S6 of the Supplementary material), the pooled results for Brazil and its five regions were similar when using a quadratic B-spline function for fitting the exposure-response relationship; equally spaced knots for temperature, maximum 7, 14 and 28-day lags; 2 and 4 equally spaced knots for both temperature and logarithm of maximum lag; 2, 4 and 8 degrees of freedom for the spline of the variable time and the model without mean relative humidity.

4. Discussion

In the present study, we investigated the total effect of temperature on mortality due to cardiovascular diseases in 27 Brazilian cities, and the pooled effect for Brazil and its 5 regions. We found that increased cardiovascular mortality was associated with low and high temperatures in most of the cities, Brazil and the Central-West, North, South, and Southeast regions. The nature of this relationship and minimum mortality temperature differed among locations.

Recent literature reviews provide evidence for effects of cold and heat in increasing cardiovascular mortality, supported by epidemiological studies conducted at different places, where cold effects appear stronger than heat effects (Giorgini et al., 2017; Liu et al., 2015; Moghadamnia et al., 2017). Similar results have been found in studies using distributed lag non-linear models (DLNM) combined with time series to investigate the effects of temperature on total cardiovascular mortality (C. Yang et al., 2015; J. Yang et al., 2015) and cerebrovascular diseases (Zhang et al., 2014) in different cities in China, and on hospitalization for coronary heart disease, myocardial infarction, stroke (Bai et al., 2018) and diabetes (Bai et al., 2016) in Canada.

Multiple physiological mechanisms may underlie the effects of ambient temperature on cardiovascular mortality. (Liu et al., 2015). These mechanisms have not been fully elucidated, but may involve vascular changes, responses of the autonomous nervous system, arrhythmias, oxidative stress and inflammation (Giorgini et al., 2017; Zhang et al., 2018). Exposure to low temperatures can lead to vasoconstriction, polyuria and increase in blood viscosity and hemoconcentration. It is also related to the production and release of inflammatory markers in the

Table 2
Relative risks and 95% confidence intervals for the effects of extremely low temperatures (1st percentile versus minimum mortality percentile) and extremely high temperatures (99th percentile versus minimum mortality percentile) on daily cardiovascular mortality, cumulated over lag 0–21 days in 27 Brazilian cities and the pooled effects for the country and its five regions.

Regions/cities	MMT (°C)	MMP	Relative risk and 95% confidence intervals	
			1st percentile	99th percentile
Brazil	27.7	0.79	**1.26 (1.17–1.35)**	**1.07 (1.01–1.13)**
Central-West	28.4	0.90	**1.46 (1.31–1.62)**	**1.13 (1.04–1.22)**
Brasília	25.5	0.90	**1.44 (1.26–1.65)**	**1.15 (1.02–1.29)**
Campo Grande	27.5	0.81	**1.60 (1.29–1.97)**	1.15 (0.93–1.41)
Cuiabá	31.5	0.94	1.27 (0.95–1.69)	1.09 (0.86–1.4)
Goiania	29.0	0.90	**1.36 (1.16–1.61)**	**1.17 (1.03–1.33)**
North	29.6	0.90	1.13 (0.95–1.35)	1.04 (0.96–1.13)
Belém	23.2	0.00	1.28 (0.73–2.22)	1.32 (0.62–2.81)
Boa Vista	34.4	1.00	4.48 (0.10–210.34)	6.61 (0.2–218.71)
Macapá	23.6	0.00	1.29 (0.64–2.62)	1.33 (0.40–4.48)
Manaus	30.6	0.89	**1.46 (1.16–1.84)**	1.06 (0.90–1.26)
Palmas	25.4	0.16	**3.78 (1.5–9.53)**	**1.75 (1.05–2.93)**
Porto Velho	28.5	0.91	1.07 (0.79–1.44)	**1.57 (1.10–2.22)**
Rio Branco	12.5	0.00	5.14 (0.88–29.99)	3.98 (0.48–33.12)
Northeast	29.15	0.90	1.07 (0.99–1.16)	0.99 (0.92–1.06)
Aracaju	30.6	1.00	1.23 (0.63–2.41)	1.18 (0.73–1.91)
Fortaleza	27.0	0.15	1.19 (0.95–1.49)	**1.22 (1.05–1.41)**
João Pessoa	30.4	1.00	1.32 (0.95–1.82)	1.10 (0.95–1.28)
Maceió	31.0	1.00	2.42 (0.97–6.00)	1.80 (0.82–3.93)
Natal	30.2	1.00	1.34 (0.84–2.14)	1.10 (0.80–1.52)
Recife	30.4	1.00	**1.36 (1.05–1.76)**	1.14 (0.96–1.35)
Salvador	31.0	1.00	1.03 (0.75–1.42)	1.02 (0.80–1.31)
São Luís	29.6	0.94	1.15 (0.82–1.60)	1.02 (0.59–1.77)
Teresina	23.0	0.00	1.16 (0.81–1.67)	1.37 (0.80–2.33)
South	25.2	0.85	**1.71 (1.55–1.90)**	1.18 (0.82–1.70)
Curitiba	29.0	1.00	**2.15 (1.47–3.16)**	1.13 (0.89–1.43)
Florianópolis	32.0	1.00	1.59 (0.61–4.13)	1.13 (0.55–2.31)
Porto Alegre	26.5	0.86	**1.88 (1.57–2.25)**	**1.56 (1.39–1.76)**
Southeast	26.1	0.74	**1.45 (1.33–1.58)**	**1.20 (1.11–1.31)**
Belo Horizonte	24.5	0.69	**1.36 (1.23–1.5)**	1.11 (0.98–1.26)
Rio de Janeiro	27.0	0.63	**1.40 (1.31–1.51)**	**1.39 (1.29–1.51)**
São Paulo	25.0	0.84	**1.60 (1.51–1.71)**	**1.15 (1.09–1.22)**
Vitória	26.5	0.62	**1.46 (1.08–1.99)**	1.24 (0.95–1.64)

MMT: Minimum mortality temperature. MMP: minimum mortality percentile.
Bold indicates the 95% CI that not include the null value.

bloodstream. Besides, the blood cholesterol level is usually higher during colder seasons. These mechanisms can lead to an increase in vascular resistance and blood pressure, coronary spasms, plaque formation and rupture, and even thrombosis, which can cause cardiovascular events (Bhatnagar, 2017; Lin et al., 2013). Exposure to heat induces sweating and increased blood volume in cutaneous regions, leading to dehydration, higher blood viscosity and hypotension, while also increasing cardiac overload (Liu et al., 2015; Luan et al., 2017).

Most of the cities and regions included in the present study exhibited non-linear exposure-response relationships consistent with a U-shaped curve, indicating higher risk above and below the minimum mortality temperature, being most marked at extreme cold and hot temperatures. A study that included 18 Brazilian cities and used DLNM models combined with generalized linear models also found a U-shape pattern for general mortality (Guo et al., 2014), and likewise for cardiovascular mortality in studies conducted in the city of São Paulo that adopted other statistical approaches (Gouveia et al., 2003; McMichael et al., 2008; Pinheiro et al., 2014; Son et al., 2016). The current results suggest no effect of cold in some cities, such as Belém, Macapá, Rio Branco and Teresina, which had minimum mortality percentiles close to zero and increased risk above this value. Other cities (Boa Vista, Aracaju, João Pessoa, Maceió, Natal, Recife, Curitiba and Florianópolis) had unexpected curves, with minimum mortality percentiles close to 1, and a tendency for a protective effect above this value.

These unexpected patterns, particularly in the Northeast region, might be associated with the narrow temperature range and low daily death count evenly distributed throughout the year. The instability of estimates of the exposure-response relationship when data are sparse and where the temperature range is very narrow was reported in a study involving 12 cities of middle and low-income countries with different climatic characteristics (McMichael et al., 2008). As observed in previous studies, locations with higher death count, more stable series and greater contrast and variability of exposure had more accurate estimates of temperature and minimum mortality and the exposure-response relationship (Gasparrini et al., 2015; Vicedo-cabrera et al., 2019).

Measurement error is another important issue when studying exposure-response relationships (Gasparrini and Armstrong, 2010; Rhomberg et al., 2011). Exposure misclassification can lead to bias in parameter estimates and mask features in both linear and nonlinear regression curves, as well as the loss of statistical power (Rhomberg et al., 2011). Furthermore, there is some evidence of the harvesting effect associated with heat in the Northeast region, roughly between the 4th and 14th lag days, and in the Southern regions, between the 1st and 3rd lag days (Fig. 3). Given this phenomenon raised risk ratio at short lags followed by an apparently protective effect (Bhaskaran et al., 2013), it may be associated with the reduction in risk at high temperatures.

The overall minimum mortality temperature for Brazil was 27.7 °C (79th percentile of the distribution of mean temperature), ranging from 25.2 °C in the South (85th percentile) to 29.6 °C in the North (90th percentile), the coldest and hottest regions, respectively. Higher temperature thresholds in a northerly direction and lower in a southerly direction suggest that people from typically hot places are more vulnerable to low temperatures, whereas those from colder places are more vulnerable to hot temperatures. Consistent with other studies, this behavior indicates that people may adapt to the climate where they live through physiological and behavioral mechanisms (Guo et al., 2014; McMichael et al., 2008; Moghadamnia et al., 2017; J. Yang et al., 2015).

The 27 cities included in the present study have different geographic, socioeconomic, demographic and infrastructure characteristics. The variable that best explained the effect heterogeneity among these cities was mean annual temperature range, where the effect of temperature was greater in places with more extensive temperature ranges. Also, places with lower latitudes and better socioeconomic conditions showed a greater temperature effect. This result may seem counterintuitive in that the temperature effect seems greater in locations with more deprived socioeconomic situations. However, the meta-variables are highly correlated, and consequently, the places with broader temperature ranges had lower latitudes and better socioeconomic situations. The temperature effects were lower for locations with narrower temperature ranges and worse socioeconomic conditions.

Our study has some additional limitations. First, because the temperature data were collected from meteorological stations and constitute a single measure for the city, there is a possibility of exposure misclassification. As discussed previously, exposure measurement error can affect parameter estimates and change the apparent shape of the relationship, particularly in nonlinear models (Rhomberg et al., 2011). Although, one previous study has shown similar results using a single exposure measure for the city and spatiotemporal model for temperature (Guo et al., 2013), the impacts of exposure misclassification error on the current study are complicated to predict.

In the present study, we used causal diagrams and distributed lag regression models for assessment and adjustment of confounding. Although these methods are well-developed, we cannot exclude the possibility of residual confounding induced by model specifications, such as the degree of smoothing (Gasparrini and Armstrong, 2010) and by unobserved time-dependent variables. However, sensitivity analyses indicate that our findings were robust under different model specifications. Also, we could not control for the effects of air pollution and infection diseases, such as influenza epidemics. However, it seems more plausible that these variables act as mediators on the causal pathways between temperature and cardiovascular mortality (Buckley et al.,

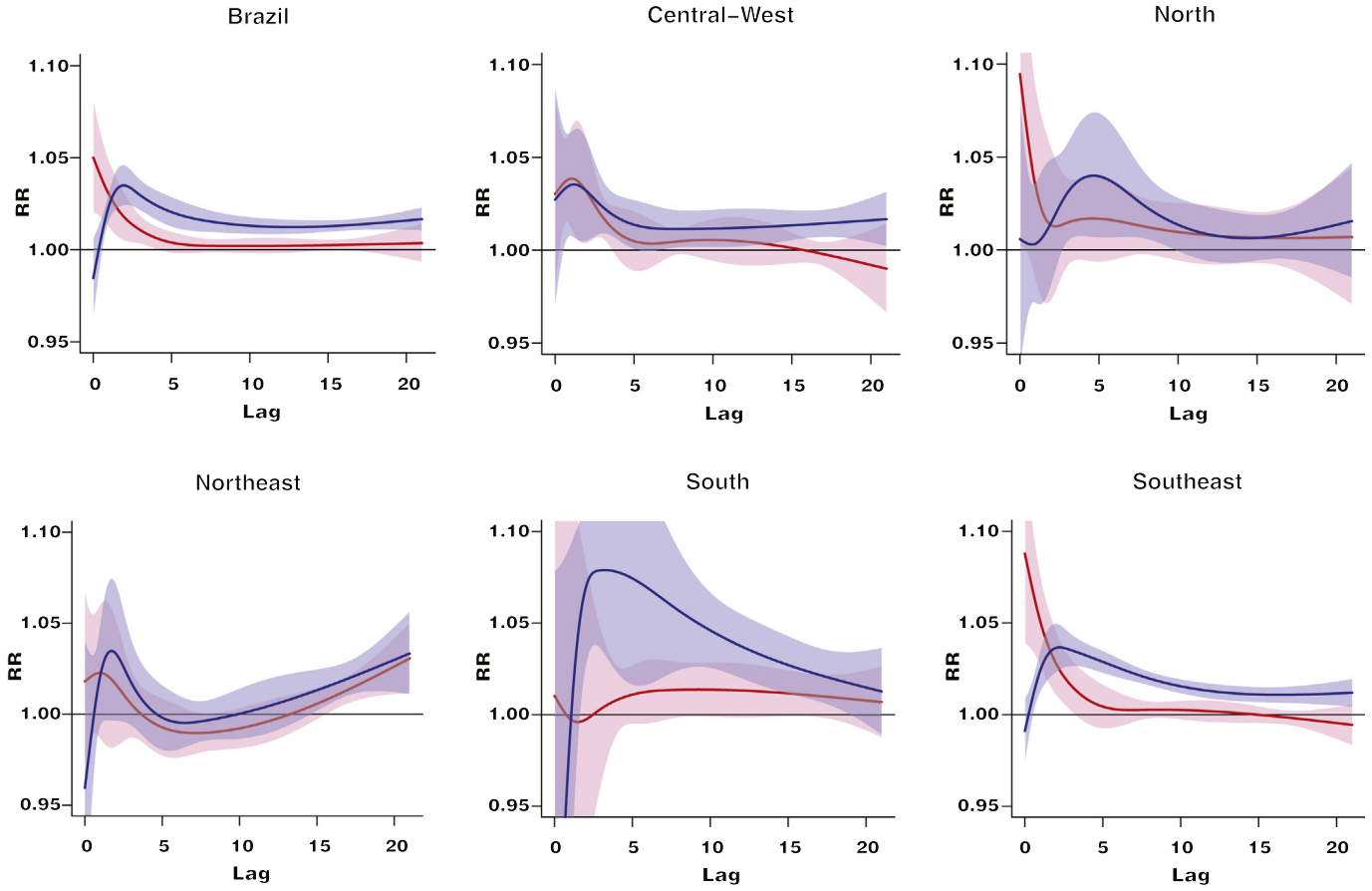


Fig. 3. Pooled lag-response relationship for low temperature (1st percentile versus minimum mortality percentile, represented in blue) and high temperature (99th percentile versus minimum mortality percentile, represented in red) associated with cardiovascular mortality, in Brazil and its five regions. The shaded blue and red areas represent the 95% confidence intervals.

2014; Imai et al., 2016), as opposed to confounding variables and should not be controlled for when estimating total effects. Finally, although the cities included in the study encompassed almost a quarter of the national population, the results pertain to urban areas and may not be representative of rural regions.

To our knowledge, this is the first study to examine the relationship between temperature and cardiovascular mortality for all the Brazilian state capitals, representing a population of over 45 million and locations with different climatic and socioeconomic characteristics. The study results add to the body of evidence about temperature effects on health, particularly on cardiovascular mortality, a topic less explored in low and middle income countries and in places with a predominantly tropical climate. Further, given the high rate of CVD and the context of climate change with rising temperatures and frequency of extreme cold and heat events (IPCC, 2014), this type of study is important to inform public policies for controlling and mitigating the impact of climate on health.

Table 3
I^2 statistic, Cochran Q test, Wald test and Akaike's information criteria (AIC) for meta-regression models.

	I^2 (%)	P-value for Q test	P-value for Wald test	AIC
Intercept-only	76.3	0.000	NA	85.8
Univariate models				
Population	71.7	0.000	0.423	226.9
Urban population	76.5	0.000	0.206	73.2
Daily deaths	71.1	0.000	0.313	129.5
Gini	73.6	0.000	0.078	75.6
Unemployment	61.8	0.000	0.000	102.4
Human development index	66.0	0.000	0.002	71.3
Average income	62.2	0.000	0.000	140.3
Living in extreme poverty	59.2	0.000	0.000	94.0
Minimum temperature	55.4	0.000	0.000	93.8
Maximum temperature	72.1	0.000	0.002	105.7
Mean temperature	61.1	0.000	0.000	95.3
Mean temperature range	53.6	0.000	0.000	95.9
Primary health care	76.0	0.000	0.738	134.6
Domiciles with trees	72.9	0.000	0.422	131.2
Latitude	60.4	0.000	0.000	107.3
Population aged 65 and above	70.8	0.000	0.000	99.7

5. Conclusions

The results of this study indicate effects of low and high temperatures on the increased risk of mortality from CVD in most of Brazil's capital cities, besides the pooled effect for Brazil and the Central-West, North, South, and Southeast regions. In general, the minimum mortality temperature was higher in hotter locations and lower in colder places. Heat effects were more immediate and persisted for only a few days, whereas cold effects were more delayed and prolonged. The temperature range was the variable that best explained the heterogeneity in the association among the 27 cities studied. Cold and heat effects were greatest in places with broader temperature range, yet low or negligible in locations with narrow temperature ranges. The study results can help inform public policies to tackle extreme heat and cold events, especially in the context of climate changes.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2019.06.493.

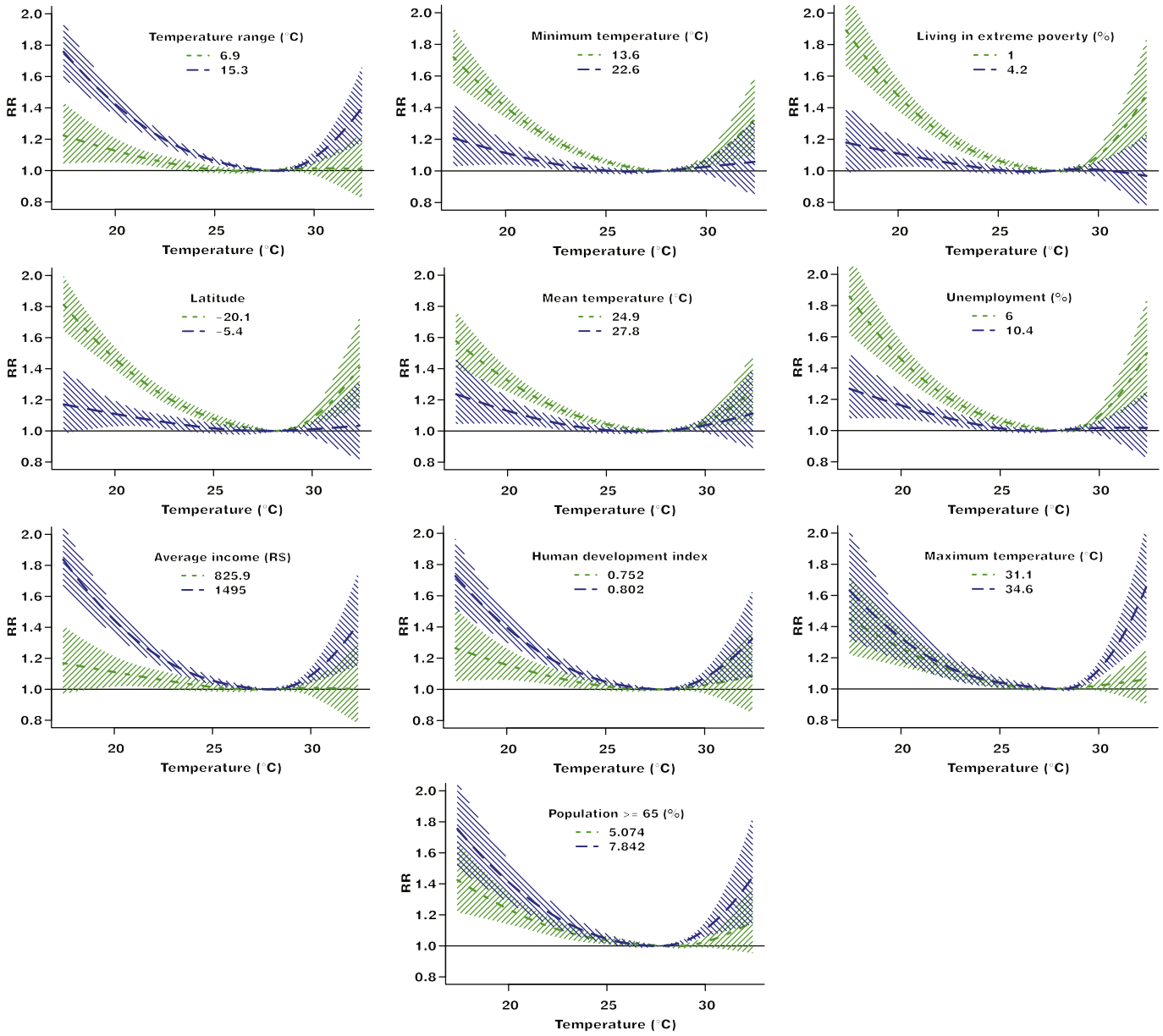


Fig. 4. Effect of temperature on cardiovascular mortality versus minimum mortality temperature for Brazil, cumulated over 21-day period, according to 25th and 75th percentiles of meta-variables temperature range, minimum temperature, percentage living in extreme poverty, latitude, mean temperature, percentage unemployment, average income, human development index and maximum temperature.

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Declaration of Competing Interest

The authors declare no conflict of interest.

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